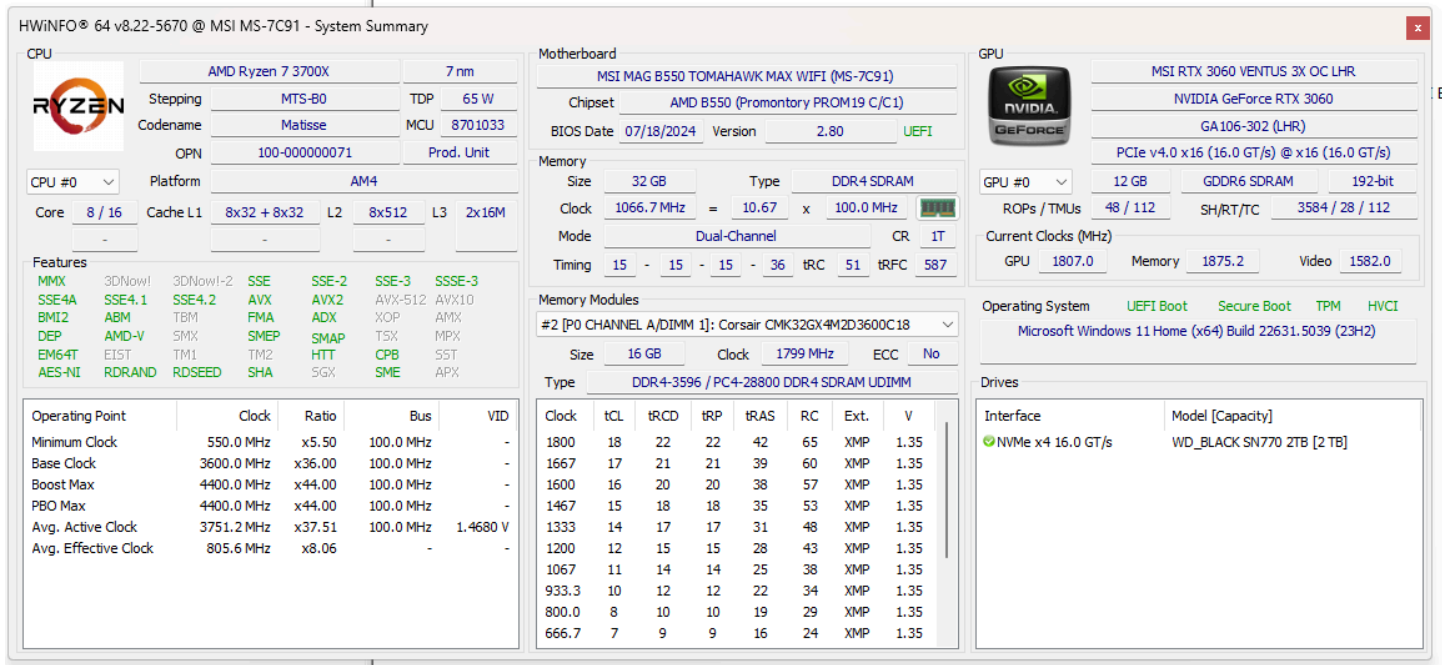
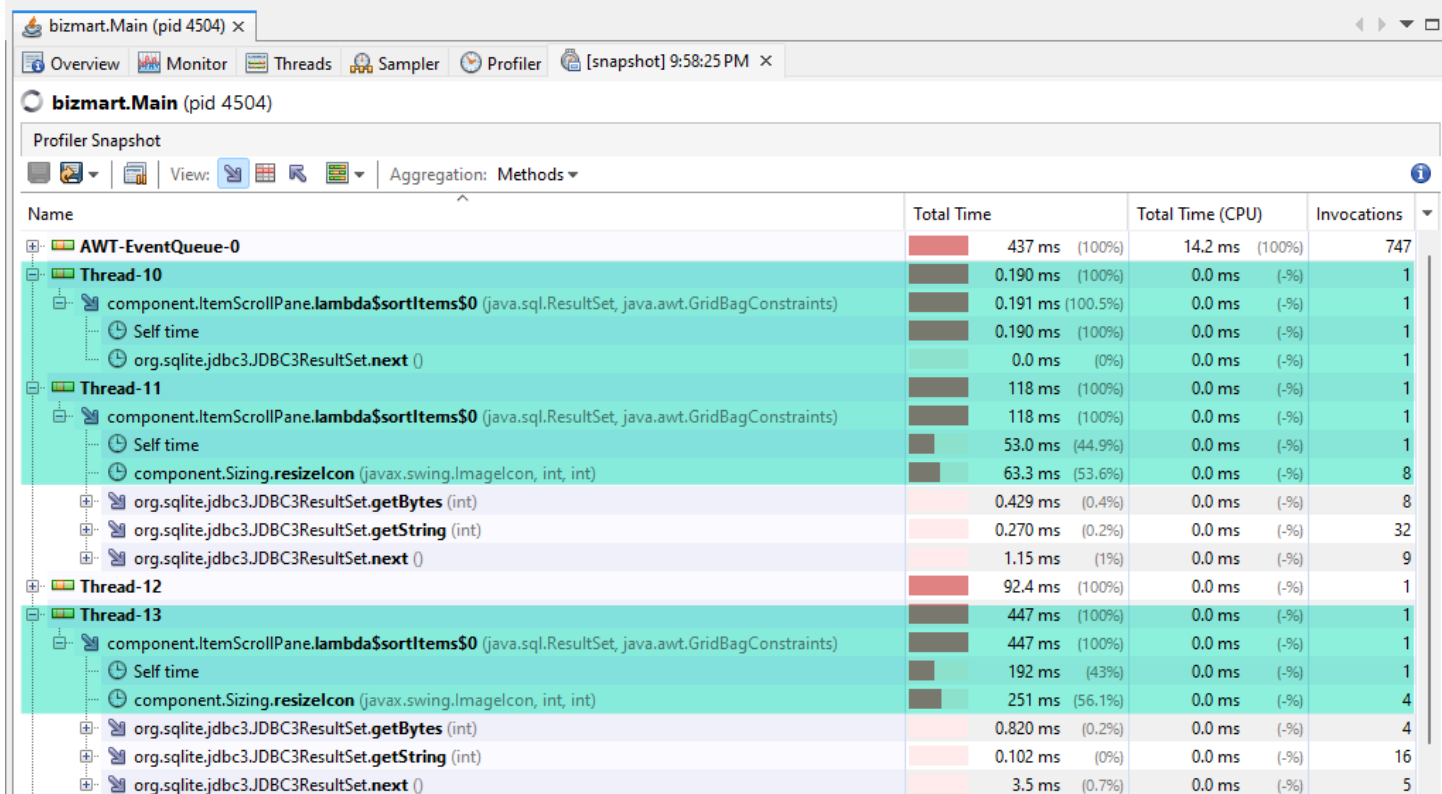


Computer Hardware

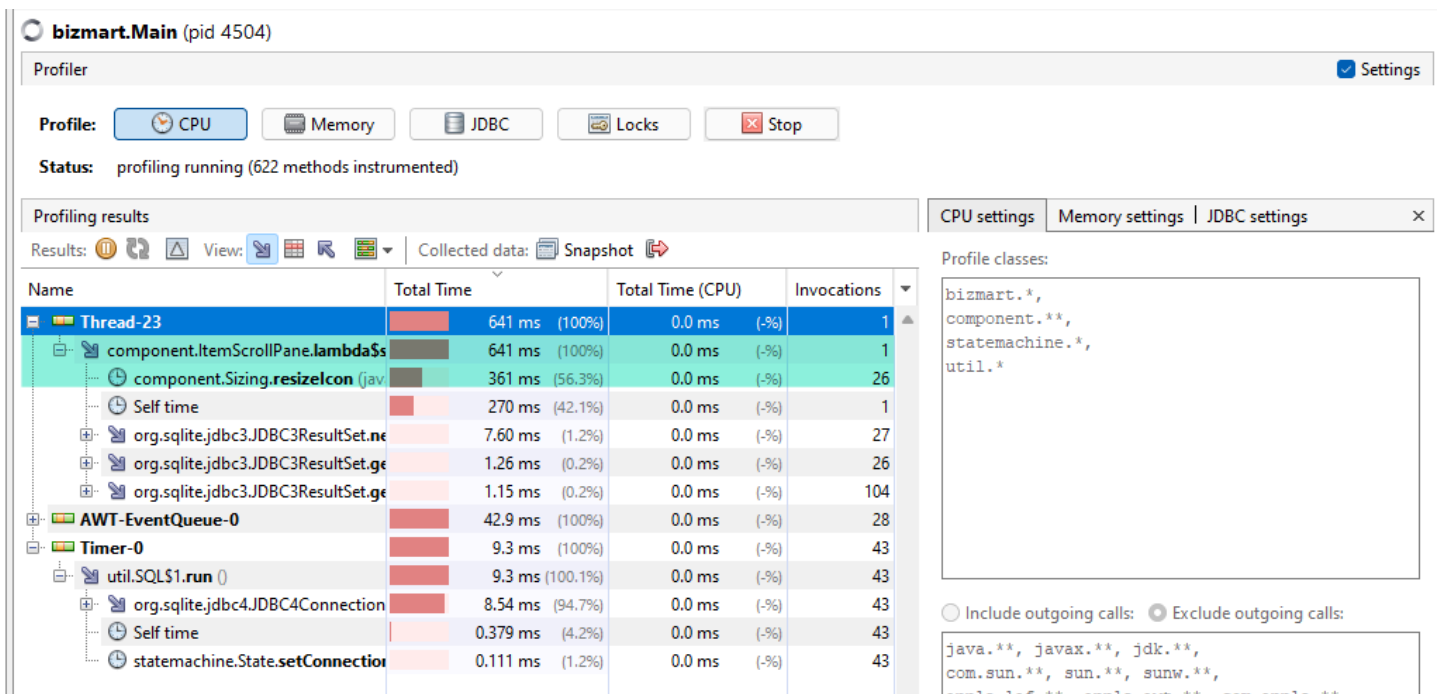


The hardware of the computer running the application during performance testing, retrieved using HWiNFO. To summarize, the PC contains an AMD Ryzen 7 CPU, an MSI B550 motherboard, and an NVIDIA GeForce RTX 3060 GPU. It also has 36 gigabytes of DDR4 ram.

Profiling CPU Usage



Through profiling the CPU usage of Bizmart threads with VisualVM, I discovered that the threads that create clickable thumbnails when searching products took significant time to execute. Right now the database is small, only 26 items, but once thousands of items are added it will take several seconds at minimum to add all of the thumbnails to the interface. Given that searching for items is the main function of the Customer state (and is also implemented by the Employee and Manager states), this issue will significantly impact the user experience if not fixed.



This slow execution is also apparent when using the application, especially when entering the Customer state where the top 100 item thumbnails are immediately added to the interface. Thread-23 in the above image took 641 milliseconds to finish creating only 26 items.

The problem is the `resizeIcon()` method, which takes over half of the execution time in each instance. Each item image in the database is at full-size, so `resizeIcon()` is used to scale the images to thumbnail size. The method essentially draws a new image each time it's called, which is inefficient and time-consuming.

However, I want to retrieve item information from the database every time it's needed, rather than saving it in the running instance of the program, because managers could change that information (including the image) at any point in time.

The solution I came up with is to call `resizeIcon()` when a Manager creates a new item or changes an item's image, then to save the thumbnail in the database. This reduces the time of the thread in two ways: 1) the smaller image file takes less time to download from the database 2) the image can be used as is and does not need to be redrawn.

bizmart.Main (pid 20264) x

Overview Monitor Threads Sampler Profiler

bizmart.Main (pid 20264)

Profiler

Profile: CPU Memory JDBC Locks Stop

Status: profiling running (512 methods instrumented)

Profiling results

Results: [Icons] View: [Icons] Collected data: Snapshot [Icon]

Name	Total Time	Total Time (CPU)	Invocations
AWT-EventQueue-0	138 ms (100%)	0.0 ms (-%)	191
Thread-12	102 ms (100%)	0.0 ms (-%)	1
component.ItemScrollPane.lambda\$sortI	102 ms (100%)	0.0 ms (-%)	1
Self time	91.9 ms (89.8%)	0.0 ms (-%)	1
org.sqlite.jdbc3.JDBC3ResultSet.next	8.16 ms (8%)	0.0 ms (-%)	27
org.sqlite.jdbc3.JDBC3ResultSet.getSt	1.56 ms (1.5%)	0.0 ms (-%)	104
org.sqlite.jdbc3.JDBC3ResultSet.getBy	0.685 ms (0.7%)	0.0 ms (-%)	26
Thread-14	43.1 ms (100%)	0.0 ms (-%)	1
component.ItemScrollPane.lambda\$sortI	43.1 ms (100%)	0.0 ms (-%)	1
Self time	36.3 ms (84.3%)	0.0 ms (-%)	1
org.sqlite.jdbc3.JDBC3ResultSet.next	6.0 ms (13.9%)	0.0 ms (-%)	27
org.sqlite.jdbc3.JDBC3ResultSet.getSt	0.623 ms (1.4%)	0.0 ms (-%)	104
org.sqlite.jdbc3.JDBC3ResultSet.getBy	0.126 ms (0.3%)	0.0 ms (-%)	26

CPU setting

Profile class

bizmart.

After making these changes, the execution time for the thread was a fourth of what it was prior.

Profiling SQL Queries

bizmart.Main (pid 4504) ×			
Overview Monitor Threads Sampler Profiler [snapshot] 10:16:53 PM ×			
bizmart.Main (pid 4504)			
Profiler Snapshot			
SQL Query			
SQL Query	Total Time	Invocations	
INSERT INTO Login (PersonID, Username, Password, Question1, Question2, Question3, Answer1, Answer2, Answer3, PositionID) VALUES ('spring25', '10% off entire cart. Ends April 2025;', 0, 0, 0, 1, '2025-03-31', '2025-04-30')	2.97 ms (16.9%)	1	
INSERT INTO Person (FirstName, LastName, AddressLine1, City, State, Zip, Email) VALUES ('VERONICA', 'STEWART', '123 ADDRESS STREET', 'CELADON', 'CA', '79845', 'veronica@gmail.com')	2.91 ms (16.5%)	1	
INSERT INTO Inventory (ItemName, ItemDescription, CategoryID, SubcategoryID, RetailPrice, WholesalePrice, Quantity, RestockThreshold, ItemImage, MaxQuantity) VALUES ('Test', 'An item to test the SQL insert statement.', 1, 1, 12.99, 3.51, 1, 1, 12.99, 3.51, 1)	2.78 ms (15.8%)	1	
UPDATE Login SET AccountDisabled = 0 WHERE LoginID = 13	2.25 ms (12.8%)	1	
UPDATE Login SET PositionID = 1 WHERE PersonID = 1013	1.79 ms (10.2%)	1	
SELECT PersonID, Username, Password, Question1, Question2, Question3, Answer1, Answer2, Answer3, PositionID FROM Login WHERE Username = 'mjane01' COLLATE NOCASE AND Password = 'sdvzvzug-4' AND (AccountDisabled = 0 OR Disabled IS NULL)	1.69 ms (9.6%)	1	
SELECT PersonID, Username, Password, Question1, Question2, Question3, Answer1, Answer2, Answer3, PositionID FROM Login WHERE Username = 'mjane01' COLLATE NOCASE AND Password = 'sdvzvzug-4' AND (AccountDisabled = 0 OR Disabled IS NULL)	0.826 ms (4.7%)	1	
SELECT * FROM Inventory WHERE InventoryID = 1004	0.279 ms (1.6%)	1	
SELECT * FROM Inventory WHERE InventoryID = 1017	0.152 ms (0.9%)	1	
SELECT ItemName, RetailPrice, Quantity, ItemImage, InventoryID FROM Inventory WHERE CategoryID = 1 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100	0.147 ms (0.8%)	1	
SELECT ItemName, RetailPrice, Quantity, ItemImage, InventoryID FROM Inventory WHERE CategoryID = 2 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100	0.140 ms (0.8%)	1	
SELECT CategoryID FROM Categories WHERE CategoryName = 'Shipping Supplies'	0.127 ms (0.7%)	4	
SELECT Email FROM Person	0.113 ms (0.6%)	2	
SELECT * FROM Inventory WHERE InventoryID = 1000	0.107 ms (0.6%)	1	
SELECT * FROM Inventory WHERE InventoryID = 1005	0.104 ms (0.6%)	1	
SELECT InventoryID, ItemName, RetailPrice, Quantity, RestockThreshold, Discontinued FROM Inventory WHERE Quantity <= RestockThreshold AND (Discontinued = 0 OR Discontinued IS NULL)	0.096 ms (0.5%)	2	
SELECT PersonID, Username, Password, Question1, Question2, Question3, Answer1, Answer2, Answer3, PositionID FROM Login WHERE Username = 'martha20' COLLATE NOCASE AND Password = 'sdvzvzug-4' AND (AccountDisabled = 0 OR Disabled IS NULL)	0.093 ms (0.5%)	2	
SELECT OrderID, PersonID, EmployeeID, OrderDate, OrderTotal FROM Orders WHERE strftime('%m', OrderDate) = '03' AND strftime('%Y', OrderDate) = '2025' UNION SELECT Null, Null, Null, Null, SUM(OrderTotal) AS Summary FROM Orders WHERE strftime('%m', OrderDate) = '03' AND strftime('%Y', OrderDate) = '2025'	0.092 ms (0.5%)	1	
UPDATE Person SET FirstName = 'VERONICA', LastName = 'STEWART', AddressLine1 = '123 ADDRESS STREET', Zip = '79845', City = 'CELADON' WHERE PersonID = 1013	0.088 ms (0.5%)	1	
SELECT DISTINCT PersonID, Login.AccountDisabled FROM Person INNER JOIN Login ON Person.PersonID = Login.PersonID WHERE Email = 'veronica' COLLATE NOCASE OR FirstName = 'veronica' COLLATE NOCASE OR LastName = 'veronica' COLLATE NOCASE	0.086 ms (0.5%)	1	
SELECT * FROM Discounts WHERE DiscountCode = 'spring25' AND (Disabled = 0 OR Disabled IS NULL)	0.085 ms (0.5%)	1	
SELECT Username FROM Login	0.078 ms (0.4%)	2	
SELECT ItemName, RetailPrice, Quantity, ItemImage, InventoryID FROM Inventory WHERE ItemName LIKE '%paper%' AND CategoryID = 1 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100	0.076 ms (0.4%)	1	
SELECT CategoryID FROM Categories WHERE CategoryName = 'Forms & Planners'	0.070 ms (0.4%)	2	
SELECT OrderID, PersonID, EmployeeID, OrderDate, OrderTotal FROM Orders WHERE strftime('%Y', OrderDate) = '2025' UNION SELECT Null, Null, Null, Null, SUM(OrderTotal) AS Summary FROM Orders WHERE strftime('%Y', OrderDate) = '2025'	0.070 ms (0.4%)	1	
SELECT * FROM Login WHERE PersonID = 1013	0.050 ms (0.3%)	1	
SELECT CategoryID FROM Categories WHERE CategoryName = 'Ink & Toner'	0.047 ms (0.3%)	2	
SELECT ItemName, RetailPrice, Quantity, ItemImage, InventoryID FROM Inventory WHERE CategoryID = 4 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100	0.043 ms (0.2%)	1	
SELECT Question1, Question3 FROM Login WHERE Username = 'martha20' COLLATE NOCASE	0.043 ms (0.2%)	1	
SELECT StoreOrderDetails.InventoryID, StoreOrderDetails.Quantity FROM StoreOrderDetails INNER JOIN StoreOrders ON StoreOrders.OrderID = StoreOrderDetails.OrderID WHERE StoreOrders.Received IS NULL OR StoreOrders.Received = 0	0.038 ms (0.2%)	2	
SELECT SubcategoryID FROM Subcategories WHERE SubcategoryName = 'Envelopes'	0.038 ms (0.2%)	1	
SELECT PersonID FROM Person WHERE Email = 'veronica@gmail.com' COLLATE NOCASE	0.029 ms (0.2%)	1	
SELECT SubcategoryName FROM Subcategories WHERE CategoryID = 1	0.025 ms (0.1%)	2	
SELECT SubcategoryName FROM Subcategories WHERE CategoryID = 4	0.015 ms (0.1%)	1	
SELECT Answer1, Answer2, Answer3 FROM Login WHERE Username = 'martha20' COLLATE NOCASE	0.013 ms (0.1%)	1	
SELECT QuestionPrompt FROM SecurityQuestions WHERE QuestionID BETWEEN 1 AND 3	0.011 ms (0.1%)	1	

The SQL queries used are about as efficient as they can be. Columns are only retrieved as needed, and data is stored in ArrayLists or other Objects when applicable.

However, you can see that any queries retrieving inventory item data are near the top of the list. This, I believe to be because of the size of the image blobs. Adding a separate column with smaller, thumbnail sized images reduced the execution time of statements where the full-sized image is not needed.

SELECT * FROM Inventory WHERE InventoryID = 1004	0.279 ms (1.6%)	1
SELECT * FROM Inventory WHERE InventoryID = 1017	0.152 ms (0.9%)	1
SELECT ItemName, RetailPrice, Quantity, ItemImage, InventoryID FROM Inventory WHERE CategoryID = 1 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100	0.147 ms (0.8%)	1
SELECT ItemName, RetailPrice, Quantity, ItemImage, InventoryID FROM Inventory WHERE CategoryID = 2 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100	0.140 ms (0.8%)	1
SELECT CategoryID FROM Categories WHERE CategoryName = 'Shipping Supplies'	0.127 ms (0.7%)	4

The execution time for the original query that gets all items with the CategoryID “1” is 0.147ms.

SQL Query		Total Time	Invocations
SELECT CategoryID FROM Categories WHERE CategoryName = 'Shipping Supplies'		0.523 ms (78.3%)	2
SELECT ItemName, RetailPrice, Quantity, ItemThumbnail, InventoryID FROM Inventory WHERE CategoryID = 1 AND (Discontinued = 0 OR Discontinued IS NULL) LIMIT 100		0.093 ms (13.9%)	1
SELECT SubcategoryName FROM Subcategories WHERE CategoryID = 1		0.052 ms (7.8%)	1

The execution time of the same query when retrieving the smaller file is 0.093ms.

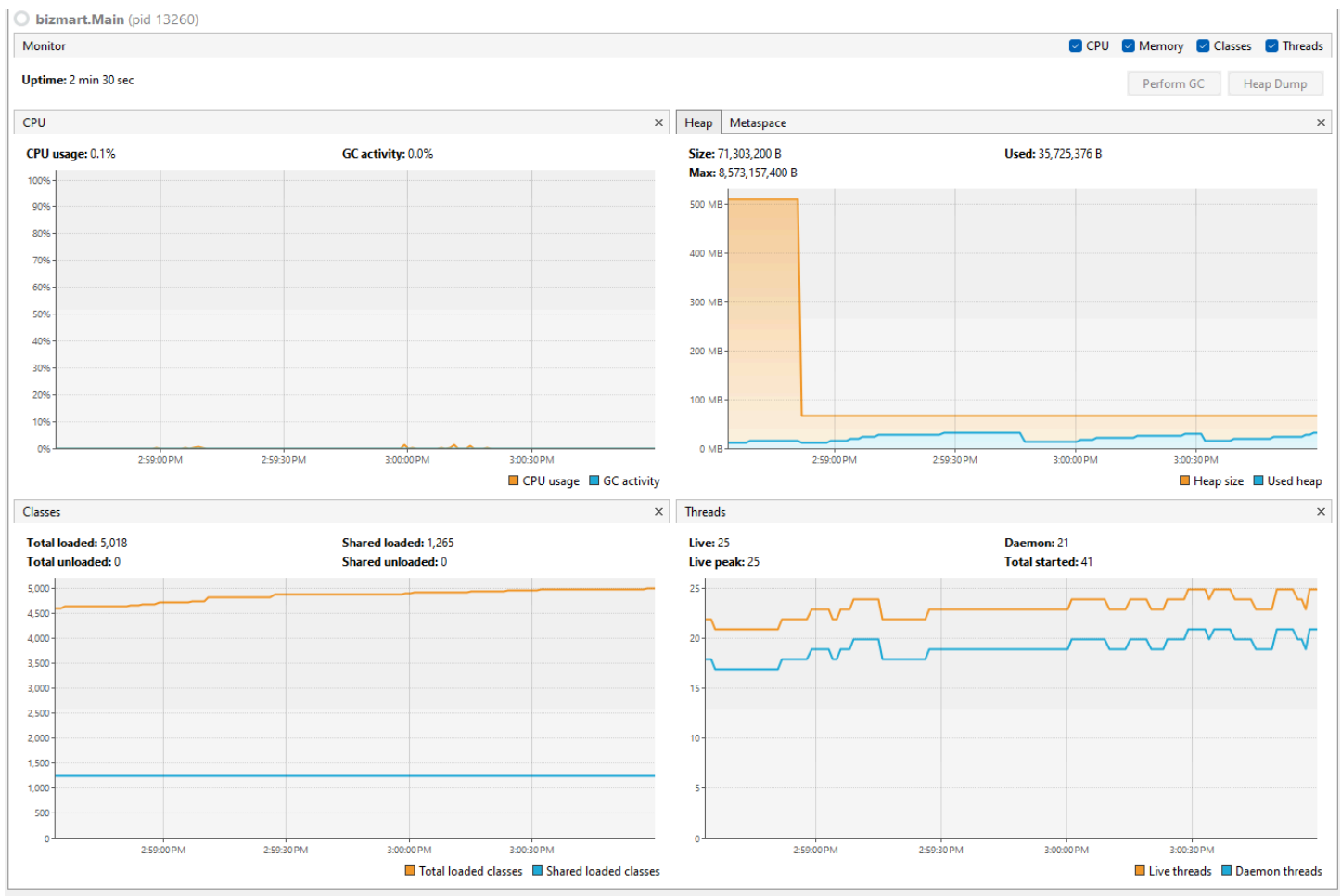
General Optimization

Using a [list](#) of tips to improve performance, I identified a few other areas for improvement.

Throughout the program there are many loops that get the size of an array within the loop. I replaced all instances of this with a variable storing the array's size, so the same value isn't being retrieved with every iteration of the loop. To locate all instances of this, I searched for `.length` and `.size` using the finder feature of Apache Netbeans in each class. I made changes to the classes Checkout and Manager.

The HTML class in the "util" package contains String objects that are concatenated multiple times, sometimes in a loop. I replaced these Strings with a StringBuilder instead to improve performance. The SQL class also uses String concatenation, but in most instances the concatenation is contained in a single statement so [the compiler automatically uses StringBuilder](#). I left the SQL class as is for this reason.

Final Results



I did a quick run through of the program after completing my optimization checklist and I didn't see anything that appeared to be a cause for concern.